

Enhancing infiltration characteristics of compact soil in open-pit dumps through arbuscular mycorrhizal fungi inoculation in *Amorpha fruticosa*: Mechanisms and effects

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ABSTRACT

Soil infiltration performance is a critical factor in the ecological restoration of waste dumps within arid and semi-arid regions. While the significance of arbuscular mycorrhizal (AM) fungi in influencing soil structure has long been acknowledged, the precise mechanism by which these fungi affect the infiltration performance of compacted soil in waste dumps remains elusive. This study, conducted over 4 years in an open-pit dump, examined the impact of artificially inoculated AM fungi on *Amorpha fruticosa* roots, soil porosity, and soil infiltration performance. We aimed to elucidate this intricate relationship through field infiltration tests and quantitative imaging technology. Our findings revealed that the development of *A. fruticosa* root significantly enhanced the soil infiltration performance and soil pore structure in the vicinity of the roots. Moreover, inoculation with AM fungi significantly improved root density characteristics, altered root morphology, and increased the average root diameter. The synergistic effect of mycorrhiza and fungi reduced the number of soil pores while increasing their mean volume and altering their geometry. Roots played a pivotal role in soil pore connectivity, with mycorrhiza and fungi significantly increasing the equivalent diameter of connected pores and the complexity of the pore network. Furthermore, inoculation with AM fungi indirectly influenced soil macroscopic pore parameters by altering root morphology. These changes in root morphology, along with alterations in soil macroscopic pore parameters, directly enhanced soil infiltration capacity. In summary, our study demonstrated that the artificial inoculation of AM fungi in open-pit dumps within arid and semi-arid areas promoted plant root growth, enhanced soil pore development, and improved soil infiltration. These findings underscored the significance of AM fungi in ecological restoration efforts.

1. Introduction

Coal stands as the cornerstone of China's energy landscape, but its extraction comes at a cost. As the nation's energy demands soar, so does the scale of coal mining operations (Huang et al., 2015). Unfortunately, this expansion has left a significant mark on the environment, with above-ground vegetation devastated, soil properties altered, and soil structure degraded (Scanlan and Hinz, 2010; Yang et al., 2015b). To address these environmental scars, reclamation and revegetation of open-pit dumps have become imperative (Simmons et al., 2008). However, this process faces numerous challenges. The disruption of natural soil structures and seed banks within open-pit dumps poses a significant

hurdle to plant recolonization (Ballesteros et al., 2012). Additionally, nutrient and water limitations further impede vegetation recovery (Haritash et al., 2007). Compounding these challenges is the fact that most of China's open-pit coal mines are situated in arid and semi-arid regions (Yang et al., 2022). In these regions, where rainfall serves as the primary source of soil water replenishment (Wang et al., 2007), soil moisture becomes a critical limiting factor for plant growth (Wang et al., 2008). Soil infiltration capacity is the key factor affecting the replenishment of soil water by precipitation (Eldridge et al., 2015). The soil in the dump has undergone both handling and compaction processes. During transport, the soil's original pore structure is disrupted, which obstructs water movement through the profile and decreases the

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infiltration rate. Compaction further increases soil bulk density, reduces porosity, and hinders the formation of new pores, thereby exacerbating the reduction in water infiltration (Zhang et al., 2015). This, meanwhile, obstructs effective rainfall penetration into the soil and its absorption by plants. Consequently, the success of open-pit mine reclamation relies heavily on ensuring adequate water supply for plant growth (Simmons et al., 2008). As such, improving soil structure has emerged as a pivotal challenge in the ecological restoration of open-pit coal mines in arid and semi-arid areas.

Infiltration, the fascinating process through which water permeates the soil, plays a vital role in the water cycle (Cui et al., 2019; Ju et al., 2024). It's a complex dance influenced by factors such as water availability and soil permeability, directly impacting surface runoff (Ebel and Moody, 2013; Liu et al., 2011). Soil infiltration performance, the rate of water penetrating per unit area under specific conditions (Assouline, 2013), depends on various factors like soil type, texture, structure, and moisture content (Liu et al., 2019). In mining areas, soil structure is often destroyed and reconstructed, disrupting the natural pore system. The natural pore system includes macropores (pores > 1.0 mm diam) formed by plant and animal activity, which are important channels for water to move downward from the soil surface (Ma et al., 2021). Even though macropores (pores > 1.0 mm diam) represent a small fraction of total soil volume, they manage over 70 % of water flux (Watson and Luxmoore, 1986). After the soil of the dump is destroyed and compacted, the number of macropores is greatly reduced, which greatly reduces the infiltration capacity of the soil (Feng et al., 2019). To delve into these intricate soil structures, researchers have turned to advanced technology. Three-dimensional X-ray computed tomography (CT) is a non-destructive powerhouse, providing direct insights into soil pore geometry (Hamamoto et al., 2016; Helliwell et al., 2013). With X-ray CT, scientists can quantify pore networks, establish pore models, and measure characteristics like size, distribution, and tortuosity (Naveed et al., 2013; Higo et al., 2011). This quantitative analysis of soil structure offers an exciting avenue for understanding the intricate relationship between soil pore structure and water infiltration (Ju et al., 2024). Generally, the impact of soil pores on infiltration is not clearly stated.

Root growth plays a crucial role in shaping the pore structure of compacted soil within opencast coal mine dumps (Volpe et al., 2013). This process is driven by two main mechanisms. Firstly, as plant roots penetrate the soil, they create gaps at the root-soil interface, forming essential channels for water infiltration due to their strong connectivity (Ceccon et al., 2011; Tracy et al., 2011). Secondly, the decay of roots by microorganisms generates large pores, further enhancing soil porosity and improving water infiltration capacity (Zhang et al., 2016b). The morphological structure of plant roots directly influences the spatial distribution of these root channels, which are closely linked to soil water infiltration (Cui et al., 2019). Roots with different structures profoundly shape the soil environment, impacting its development from the surface to the subsurface (Ola et al., 2015; Roy and McDonald, 2015). The surface mine dump is reconstructed soil and the root system formed by artificial vegetation plays an important role in the development of soil pore system (Zhang et al., 2015). Hence, investigating the morphological transformations in plant roots and their impact on soil pore evolution and infiltration in the face of soil compaction within opencast coal mine dumps is of paramount importance.

Approximately 70 % of plant roots form symbiotic relationships with arbuscular mycorrhizal (AM) fungi, giving rise to mycorrhizae (Smith and Read, 2010). These fungi, with their slender, highly branching hyphae, play a pivotal role in shaping root morphology and altering plants' nutrient acquisition strategies (He and Zhong, 2012; Shen et al., 2020). Mycorrhizal hyphae extend vast underground networks, competing for limited resources on behalf of plants and facilitating the transfer of nutrients and water from the soil to the roots. Inoculation with AM fungi can increase the root length and root volume of *Robinia pseudoacacia* root by about 1 time (Zhang et al., 2016a). Similarly, Berta et al. (1995) also found that AM fungi colonization could increase root growth of

Prunus cerasifera by 100 %. Consequently, significant amounts of mycelium are produced by AM fungi, thereby modifying root biomass and morphology (Chen et al., 2018; Shi et al., 2011).

In arid and semi-arid regions, the natural ecological restoration of open-pit dumps is a slow process. Effective methods to expedite this restoration include land reclamation and vegetation reconstruction (Nunez-Mir et al., 2015). Both field and laboratory studies have demonstrated that reclaimed vegetation inoculated with AM fungi enhances soil nutrient levels and promotes plant growth (Diagne et al., 2020; Moreira et al., 2020). The complex hyphal network expands the range of nutrients and water absorbed by plants (Wang et al., 2022). Furthermore, mycelium and root metabolism input a large amount of organic matter into the soil (Jeffries et al., 2003). Sixteen months after the AM fungi inoculation, the rhizosphere of *Amorpha fruticosa* in the treated plots showed a 44 % increase in acidic phosphatase activity compared to the control plots, and the available phosphorus content was 2.5 times higher than that of the control (Li et al., 2015). Consequently, the application of AM fungi agents is recognized as an effective approach to facilitate the restoration of degraded ecosystems, such as deserts and waste dumps (Orlowska et al., 2010; Taheri and Bever, 2011). However, the potential mechanism of AM fungi-plant-pore interactions fungi remains unclear.

To explore the significant role of AM fungi in combination with plants in the infiltration process of opencast coal mine dumps, this study aimed to investigate the effects of AM fungi inoculation, particularly with *Amorpha fruticosa*, on the infiltration characteristics of compacted soil within the dumps. The study proposed the following hypotheses:

- (1) Inoculation with AM fungi promoted root growth and root morphological traits of plants in the dump
- (2) Plant roots development directly promotes soil porosity development and water infiltration.
- (3) The impact of AM fungi on water infiltration was primarily indirect, mediated through their influence on roots and soil pores.

2. Materials and methods

2.1. Study area

The Hedaigou Open Pit mine is situated in the eastern part of Junge Banner, Ordos City, in the northern region of the Yellow West Coast Loess Plateau, with geographical coordinates of 111°13'–111°20' E and 39°43'–39°49' N (Fig. 1). The climate in the mining area is characterized as a semi-arid continental climate within the middle temperate zone. The average annual temperature ranges from 5.3 °C to 7.6 °C, and the average annual rainfall ranges from 231.0 mm to 459.5 mm, predominantly occurring from July to September, accounting for 60 % to 70 % of the annual precipitation. The average annual evaporation ranges from 1,824.7 mm to 2,896.1 mm, and the sunshine duration is 3,119.3 h. This area is located in the northeast of the Ordos platform in the Loess Plateau, characterized by loess hilly and gully terrain. According to Chinese Soil Classification (CST), the soil type in the study area is loessial soil (CRGCST, 2001). It was a light yellow color, small specific heat capacity, large diurnal temperature variation, is slightly alkaline soil, and has low fertility. The mining area has been under construction since 1989, with each production area covering a total area of 65.46 km². This includes an area of 2.10 km² for the East dump, 1.70 km² for the West Dump, 1.98 km² for the North Dump, 55.60 km² for the Inner Dump, and 1.52 km² for the Tinwan Dump. Since the reclamation project began in 1992, all solid waste reclamation in the landfill has utilized immature deep loess.

For this research, the experimental site was selected from the Inner Dump, and the integrated process of "mining, slag discharge, and soil cover" was adopted. The thickness of the soil cover was 2 ~ 3 m, as the soil was compacted by large-scale machinery.

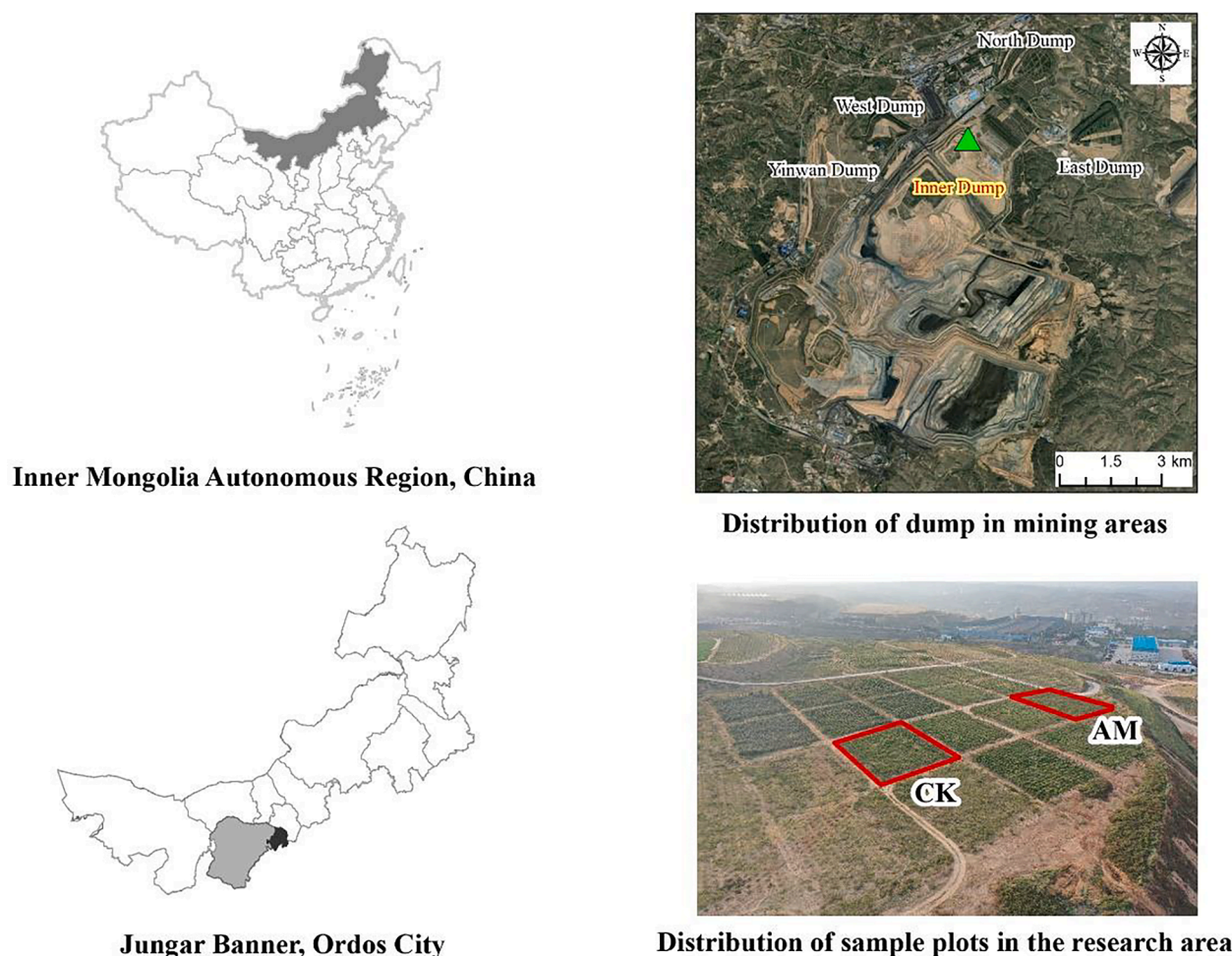


Fig. 1. Location of the study site. AM: *A. fruticosa* inoculated with AM fungus *Funneliformis mosseae*; CK: *A. fruticosa* without AM fungi inoculation.

2.2. Test plot layout

The test area was soil-covered in March 2018, and *A. fruticosa* was planted in April 2018 when the seedlings reached an average height of 50 cm. Two treatment methods were established: inoculation treatment (AM) and control treatment (CK). Each treatment was assigned to an experimental plot measuring 44 m in length and 34 m in width. *A. fruticosa* was planted with a spacing of 2×2 m. *Funneliformis mosseae* (F.m), an AM fungus, was inoculated with *A. fruticosa*. The fungus inoculant is a granular solid. During the planting process, the seedlings are first placed in the soil pit, then 50 g of the fungus inoculant is applied to the roots of the seedlings, and finally the soil is filled. Following the completion of planting in each plot, irrigation was carried out to achieve maximum saturated soil water capacity. Irrigation was conducted once a week, with continuous irrigation carried out for 1 month.

2.3. In situ double-ring infiltration experiment

In June 2022, the growth status and understory vegetation of

A. fruticosa in each plot were investigated and counted. The indicators of understory vegetation and soil physicochemical properties under different treatments were found to be similar (Table 1). Three representative sample points were selected for each treatment, with each sample point centered on *A. fruticosa* plants. Two areas were designated for infiltration tests: the proximal root area (AM-P, CK-P), located 30 cm away from the *A. fruticosa* plants, and the distal root area (AM-D, CK-D), also 30 cm away from the plants. Double-loop infiltration tests were conducted using steel tubes with different diameters, including an outer ring diameter of 40 cm and an inner ring diameter of 20 cm. Before the infiltration test, the surface vegetation of the sample site was removed, and the ground was leveled. The inner and outer rings were then driven slowly and evenly into the soil to a depth of 20 cm using a rubber mallet, minimizing interference with the soil structure. Water was added to the inner and outer rings to maintain a water head height of 15 cm. The infiltration rate was calculated by recording the change in water added to the inner and outer rings over time. The recording interval was carried out over a total of 90 min, with a denser recording frequency in the early stages and a sparser frequency in the later stages. Following the

Table 1
Basic information of sample plots.

Sample	Plant height (cm)	Abundance	Bulk density ($\text{g}\cdot\text{cm}^{-3}$)	Initial soil water content (%)	Soil organic matter ($\text{g}\cdot\text{kg}^{-1}$)	available phosphorus ($\text{mg}\cdot\text{kg}^{-1}$)	available potassium ($\text{mg}\cdot\text{kg}^{-1}$)
AM	170 ± 5	32 ± 4	1.55 ± 0.07	6.75 ± 0.5	7 ± 1.2	5.3 ± 1.8	110 ± 25
CK	110 ± 8	33 ± 3	1.55 ± 0.05	6.02 ± 0.8	6.5 ± 0.9	6.2 ± 0.9	118 ± 32

AM: *A. fruticosa* inoculated with AM fungus *Funneliformis mosseae*; CK: *A. fruticosa* without AM fungi inoculation;

infiltration test, the infiltration area was covered with a black plastic sheet and left for 24 h.

2.4. Soil sampling and measurements

Soil samples were collected from a depth of 5 to 15 cm within the inner ring of each plot. Initially, a polyvinyl chloride (PVC) ring, 5 cm in diameter and 5 cm in height, was used to extract undisturbed soil, following the procedure described by Sammartino et al. (2012). The extracted soil column was then sealed with plastic film, wrapped in sponge, and immediately transported to a laboratory for X-ray CT. After scanning, the soil-root mixture within the PVC ring was separated, and the roots were carefully collected. Following washing, root scanning was performed using a ScanMaker i800 Plus root scanner. Subsequently, the PVC ring was placed in a 60 °C oven for 72 h, and the dry mass was measured (DM, in grams). Root functional traits were categorized into two groups: density characteristics and morphological characteristics. Density characteristics included total root length density (RLD, in mm/100 cm³), root surface area density (RSD, in mm²/100 cm³), root volume density (RVD, in mm³/100 cm³), and root tip number (Tip number), which were determined through image analysis. Morphological characteristics included average root diameter (D, in mm), specific root length (SRL, in m/g), calculated as total root length divided by root dry mass, specific root surface area (SRA, in cm²/g), calculated as total surface area divided by root dry mass, and root tissue density (RTD, in mg/cm³), calculated as root dry mass divided by soil volume. Additionally, the contribution rate of the root system to pore space (root volume/pore volume) was calculated. Furthermore, approximately 250 g of disturbed soil was collected, and the mesh crossover method, as described by Abbott et al. (1984), was employed to determine the mycelium density in the soil.

2.5. CT scanning and image analysis

Undisturbed soil cores, obtained from PVC rings, underwent scanning using an ACTIS300-320/225 Industrial CT (China University of Mining and Technology, Beijing), employing X-ray source parameters set at a voltage of 100 kV and a current of 100 μA. The sample table underwent horizontal rotation at a consistent speed from 0° to 360°. Throughout the scanning process, a total of 700 detailed low-noise

images were captured, each with a resolution of 60 μm. The images were processed using Avizo 9.0 software for 3D reconstruction and visualization. To eliminate boundary effects, the middle portion of each image was extracted (Fig. 2a). Each voxel represented a volume of 60 × 60 × 60 μm, resulting in a sub-volume with dimensions of 28,200 × 28,200 × 42,000 μm for subsequent analysis (Luo et al., 2022). To establish the strength range of pore and soil matrix materials, the strength range partitioning tool was employed to automatically estimate the threshold for different material separation. Subsequently, voxels were categorized as either macropores or solids by manually adjusting the initial range threshold in the interaction threshold setting (Fig. 2b) (Hasan and Alshibli, 2010). The segmentation was based on global threshold algorithm. Following this, the pore number and the average values of pore geometry parameters (length, thickness, width, area, inclination, dip angle, shape factor, volume, and flatness) were determined through 3D reconstruction and display (Fig. 2c). Additionally, the length, thickness, width, area, inclination, dip angle and volume can be obtained directly through the three-dimensional model. Flatness is the ratio of the smallest to the medium eigenvalue of the covariance matrix. Length-width ratio is the ratio of length to width. The Shape factor is calculated by the following formula (Avizo, 2022):

$$\text{Shape factor} = \text{Area}^3 / (36 \times \pi \times \text{Volume}^2).$$

Pores were further classified into communicating pores (Fig. 2d) and independent pores (Fig. 2e). The pore fraction and pore fractal were determined accordingly. Pores fractal adopts the method of Falconer (1985). The Pores fraction is calculated by the following formula:

$$\text{Pores fraction} = \text{Volume}_{\text{pore}} / \text{Volume}_{\text{soil column}}.$$

Finally, a ball-and-stick model (Fig. 2f) was constructed for the connected pores to determine the number of pores and number of throats. The formulas for Tortuosity and EqDiameter (Equivalent diameter) are noted below (Avizo, 2022; Sebai et al., 2022):

$$\text{Tortuosity} = L_{\text{curved}} / L_{\text{chord}}$$

$$\text{EqDiameter} = \sqrt[3]{6 \times \text{Volume}_{\text{pore}} / \pi}$$

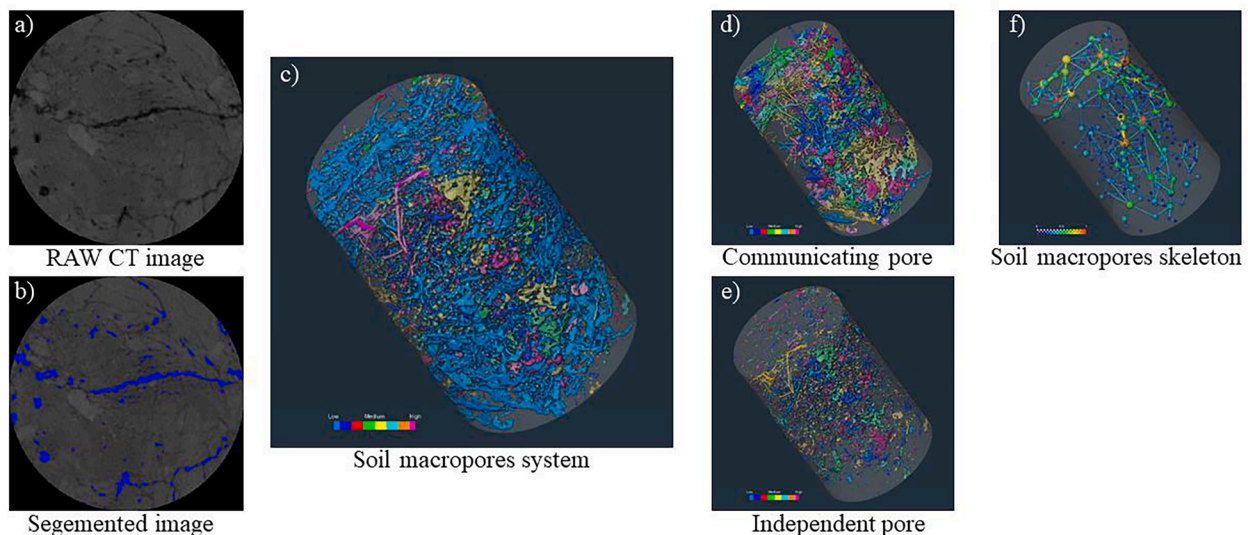


Fig. 2. Steps to capture the region of interest (ROI) and the main steps to obtain soil pore and soil pore skeleton structures. a), b) is the original image after the threshold segmentation image. c) The visual 3D map of soil holes obtained by segmentation of threshold values. d), e) are visual three-dimensional maps of connected pores and independent pores. f) The soil-connected pore skeleton structure is obtained for the automatic skeleton module in Avizo software, where solid circles of different colors represent holes of different apertures, and the short rod between the solid circles represents the throat of the holes.

2.6. Statistical analysis

IBM SPSS Statistics 26 software was employed for statistical analysis. One-way analysis of variance (ANOVA) was conducted to assess the effects of inoculation on infiltration parameters, pore parameters, root parameters, and mycelial parameters. Pearson correlation analysis was performed to determine the relationship between infiltration parameters and pore parameters, root parameters, and mycelial parameters. Origin 2021 software was used for mapping. For constructing a path model of the influence of inoculated AM fungi on infiltration characteristics, the “plsmp” package in RStudio was utilized for partial least squares path analysis.

3. Results

3.1. Characteristics of soil infiltration

Initially, there was a noticeable decrease in soil infiltration rate (0 ~ 10 min), which gradually stabilized (10 ~ 90 min) (Fig. 3a). The IIR value near the root was significantly higher than that of the distant root, with the distant root being 1.94 to 2.36 times that of the near root (Fig. 3b). Inoculation with AM fungi resulted in a significant increase of about 41 % in the IIR value near the root. The T₁₅IR value (Fig. 3c) indicated that AM-P had the highest rate (12.67 mm/min), while CK-D had the lowest rate (12.67 mm/min). Both SIR values (Fig. 3d) and AIR values (Fig. 3e) were significantly higher at the proximal end of the root than at the distal end, with the distal ends being 4.20 to 4.75 times and 1.40 to 2.46 times higher, respectively. Inoculation with AM fungi led to a significant increase of about 80 % in the SIR value near the root and about 66 % in the AIR value. The root system significantly enhanced the soil infiltration capacity of the dump, with AM fungi inoculation notably promoting soil infiltration near the root.

3.2. Characteristics of soil pores

Table 2 presents the soil pore number and pore geometry parameters. Comparison between the proximal and distal roots revealed that the

Table 2
Soil pore number and geometric characteristics.

Sample	AM-P	CK-P	AM-D	CK-D
Length/ μm	447.91 $\pm$ 70.06a	529.93 $\pm$ 107.54a	245.71 $\pm$ 12.29b	236.98 $\pm$ 12.73b
Width/ μm	218.33 $\pm$ 27.8a	255.52 $\pm$ 59.34a	129.9 $\pm$ 3.7b	124.97 $\pm$ 7.67b
Thickness/ μm	193.63 $\pm$ 20.52a	225.2 $\pm$ 54.7a	112.61 $\pm$ 4.13b	107.9 $\pm$ 7.64b
Area/ mm^2	1.74 $\pm$ 0.92a	1.19 $\pm$ 0.25ab	0.24 $\pm$ 0.11b	0.54 $\pm$ 0.37b
Volume/ mm^3	0.27 $\pm$ 0.15a	0.13 $\pm$ 0.04ab	0.01 $\pm$ 0.01b	0.04 $\pm$ 0.04b
dip angle/ $^\circ$	7.67 $\pm$ 2.98ab	5.75 $\pm$ 4.39b	13.27 $\pm$ 4.69ab	14.65 $\pm$ 4.84a
inclination/ $^\circ$	69.72 $\pm$ 3.23ab	66.74 $\pm$ 1.6b	72.73 $\pm$ 2.99a	73.71 $\pm$ 2.84a
Flatness	0.13 $\pm$ 0.03a	0.18 $\pm$ 0.07a	-0.18 $\pm$ 0.03b	-0.24 $\pm$ 0.06b
shape factor	1.76 $\pm$ 0.46ab	1.83 $\pm$ 0.17a	1.1 $\pm$ 0.24b	1.98 $\pm$ 0.46a
Length-width ratio	1.60 $\pm$ 0.03a	1.62 $\pm$ 0.04a	1.54 $\pm$ 0.01b	1.53 $\pm$ 0.02b
Pore number	13267.00 $\pm$ 6438.55b	13077.00 $\pm$ 4985.17b	91315.67 $\pm$ 17003.31a	29418.67 $\pm$ 4289.8b

AM-P, Proximal roots of *A. fruticosa* inoculated with AM fungus; CK-P, Proximal roots of *A. fruticosa* without AM fungi inoculation; AM-D, Distal roots of *A. fruticosa* inoculated with AM fungus; CK-D, Distal roots of *A. fruticosa* without AM fungi inoculation. Values with different letter are significantly different at $p < 0.05$.

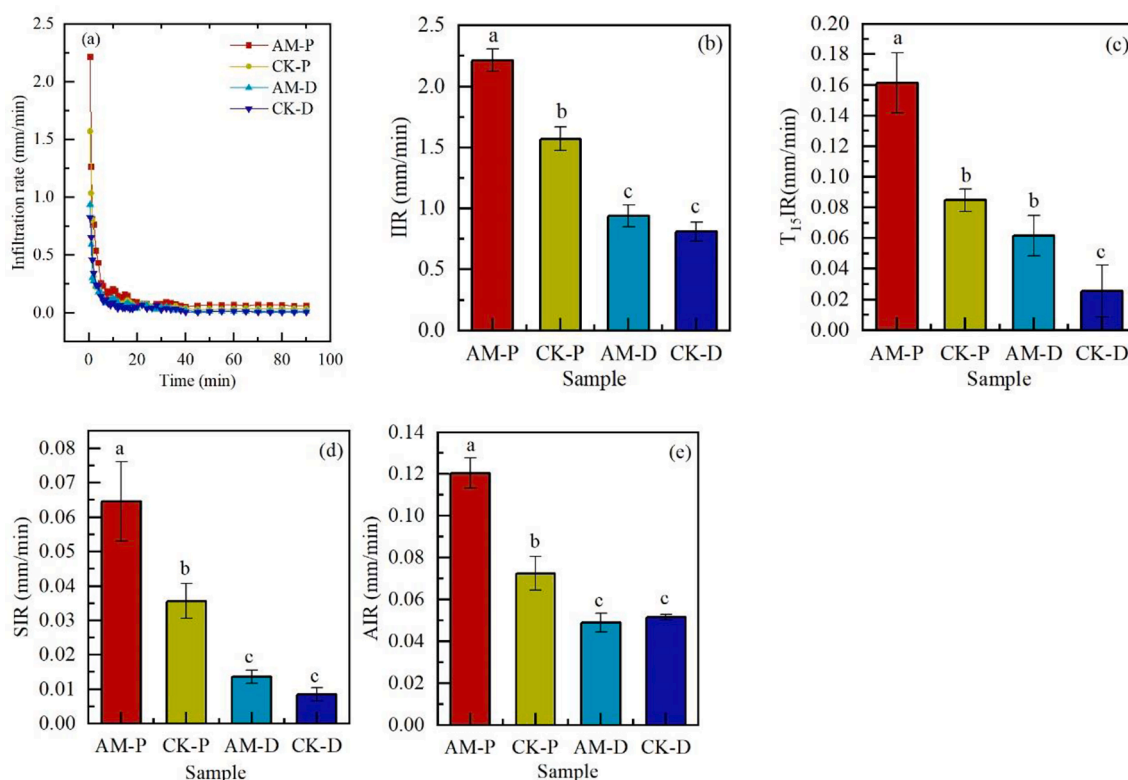


Fig. 3. AM-P, Proximal roots of *A. fruticosa* inoculated with AM fungus; CK-P, Proximal roots of *A. fruticosa* without AM fungi inoculation; AM-D, Distal roots of *A. fruticosa* inoculated with AM fungus; CK-D, Distal roots of *A. fruticosa* without AM fungi inoculation. a) Infiltration process, b) Initial infiltration rate (IIR), c) Infiltration rate of 15 min (T₁₅IR), d) Stable infiltration rate (SIR), e) Average infiltration rate (AIR). Values with different letter are significantly different at $p < 0.05$.

average pore length, width, and height of the proximal root were significantly greater than those of the distal root ($P < 0.05$). Additionally, the length, width, and height of the proximal root treated with bacteria were approximately 80 % of those in the control treatment. In terms of pore surface area (1.74 mm^2) and volume (0.27 mm^3), AM-P exhibited significantly higher values compared to other treatments ($P < 0.05$). The flatness value was positive near the root, whereas it was negative farther from the root. Furthermore, the length-width ratio of the proximal root (ranging from 1.56 to 1.66) was significantly higher than that of the distal root (ranging from 1.51 to 1.55) ($P < 0.05$). In terms of pore parameter characteristics, the inoculation effect on pore shape and position was not significant ($P > 0.05$), while the root system significantly influenced pore shape and reduced pore count.

Table 3 presents the characteristics of soil porosity and fractal dimension. The total soil porosity at the proximal root was significantly greater than that at the distal root ($P < 0.05$). Specifically, AM-P (11.01 %) had 20 times the porosity of AM-D (0.55 %), and CK-P (5.95 %) had 37 times the porosity of CK-D (0.16 %). Inoculation with AM fungi resulted in an 85 % increase in the total porosity of the proximal root. Moreover, 88 % of the total pores in AM-P and 57 % in CK-P were connected pores. The inoculation of AM fungi significantly increased the proportion of connected pores in the soil pore system ($P < 0.05$), with the connected porosity of AM-P (9.71 %) being 2.96 times that of CK-P (3.28 %). In both AM-D and CK-D soil systems, pores were independent, resulting in low porosity, approximately 0.16 % to 0.55 %. The fractal dimension of total porosity followed this order: AM-P (2.36) > CK-P (2.29) > AM-D (1.98) > CK-D (1.69). Additionally, the fractal dimension of independent pores was significantly higher at the proximal root than at the distal root ($P < 0.05$).

The characteristic parameters of connected pores are illustrated in Fig. 4. The tortuosity of connected pores was higher in CK-P (2.20) compared to AM-P (1.69). Inoculation treatment significantly increased the equivalent diameter of connected pores ($P < 0.05$), with the equivalent diameter of AM-P (16.79 mm) being 1.59 times that of CK-P (2.20 mm). Inoculation with AM fungi significantly increased the number of pore throats in the pore structure ($P < 0.05$). Specifically, the number of pores increased by 1.91 times, and the number of throats increased by 1.82 times.

3.3. Characteristics of soil roots and mycelia

Fig. 5 (a ~ d) depicts the density characteristics of plant roots in the soil. RLD, RSD, RVD, and Tip number were notably higher in the proximal root compared to the distal root ($P < 0.05$). The RVD value of AM-P ($7,296.83 \text{ mm}^3/100 \text{ cm}^3$) was 2.75 times that of CK-P ($2,648.74 \text{ mm}^3/100 \text{ cm}^3$). Fig. 5 (e ~ h) illustrates the morphological characteristics of plant roots in the soil. The root diameter (D) of AM-P (0.72 mm) was significantly greater than that of CK-P (0.49 mm), AM-D (0.42 mm), and CK-D (0.41 mm) ($P < 0.05$). Bacteria treatment increased the proximal root diameter by 46.94 %.

Regarding RTD performance, the sequence was AM-P (2.73 mg/cm^3) > CK-P (11.77 mg/cm^3) > AM-D (0.24 mg/cm^3) > CK-D (0.06 mg/cm^3). The SRL of CK-P (59.24 m/g) was notably higher than that of other treatments ($32.60 \sim 43.56 \text{ m/g}$) ($P < 0.05$). CK-P had the highest SRA

($0.09 \text{ cm}^2/\text{g}$), while AM-D had the lowest ($0.05 \text{ cm}^2/\text{g}$). The contribution rate of the root to the pore was significantly higher at the proximal root than at the distal root ($P < 0.05$). Inoculation increased the root's contribution rate to the pore by approximately 50 %. After inoculation with AM fungi, the mycorrhizal colonization (MC) (28.67 %) of AM fungi in AM-P roots (Fig. 6a) and soil hyphal density (HD) (m/g) (Fig. 6b) were significantly higher than those in other treatments ($P < 0.05$).

3.4. Mechanism of AMF on the infiltration characteristics of open-pit dump

Table S1 presents the Pearson correlation between infiltration vegetation and its influencing factors. Inoculation with AM fungi (MC and HD) and root density index (RLD, RVD, RSD, and Tips) showed significant ($P < 0.05$) or extremely significant ($P < 0.01$) positive correlation with infiltration index. Among the root morphological indexes, root diameter (D) and RTD were significantly positively correlated with the infiltration indexes ($P < 0.01$). Regarding macroscopic pore characteristics, pore fraction, number of pores, number of throats, and equivalent diameter (EqDiameter) were significantly positively correlated with infiltration indexes ($P < 0.01$). Among the pore morphology characteristics, only pore area, volume, and flatness were significantly ($P < 0.05$) or extremely significantly ($P < 0.01$) positively correlated with the infiltration indexes. The influence factors with strong correlation with the infiltration index were fitted to the infiltration index. It was found that the influence factors were significantly positively correlated with the infiltration index. Among them, MC and HD could explain 61.4 % to 78.4 % of the variation of $T_{15}\text{IR}$, SIR and AIR (Fig. 7a, b). RLD could explain 67.6 % of the variation in IIR (Fig. 7c). RVD could explain 84.1 % to 94.9 % of the variation in $T_{15}\text{IR}$, SIR, and AIR (Fig. 7d). D could explain 84.9 % to 95.6 % variation in $T_{15}\text{IR}$, SIR, and AIR (Fig. 7e). Pores fraction can account for 80 % to 88.2 % of the variation in $T_{15}\text{IR}$, SIR, and AIR (Fig. 7f). Mean pore volume could explain 64.1 % variation of IIR (Fig. 7g), and EqDiameter could explain 79.7 %-84.5 % variation of $T_{15}\text{IR}$, SIR and AIR (Fig. 7h).

Partial least squares path analysis was conducted to develop a structural equation model assessing the influence of AM fungi inoculation on soil infiltration characteristics (Fig. 8). The model exhibited a goodness of fit (GOF) of 0.838. Predictors elucidated 54.1 % (R^2) of root density changes, 75.9 % (R^2) of root morphology changes, 95.9 % (R^2) of root macroporosity changes, and 86.3 % (R^2) of pore morphology changes. Additionally, the model explained 96.4 % (R^2) of pore morphology changes and alterations in infiltration characteristics.

The analysis revealed that inoculation with AM fungi positively influenced root density and morphological characteristics ($P < 0.01$). Subsequently, root morphological characteristics positively affected macroscopic pore features and soil infiltration characteristics ($P < 0.01$). Furthermore, macroscopic pore features had a positive impact on soil infiltration characteristics ($P < 0.05$). In summary, inoculation with AM fungi indirectly influenced soil macroporosity and infiltration characteristics primarily through its effect on root morphology. Changes in root morphology directly affected soil infiltration and indirectly impacted infiltration through alterations in macroporosity.

Table 3

The pore fraction and fractal dimension of independent and continuous pores.

Sample	Pores fraction (%)			Fractal		
	Total	Independent	Continuous	Total	Independent	Continuous
AM-P	$11.01 \pm 3.09\text{a}$	$1.3 \pm 0.26\text{ab}$	$9.71 \pm 3.35\text{a}$	$2.36 \pm 0.02\text{a}$	$2.01 \pm 0.04\text{ab}$	$2.31 \pm 0.02\text{a}$
CK-P	$5.95 \pm 0.43\text{b}$	$2.5 \pm 1.89\text{a}$	$3.28 \pm 2.24\text{b}$	$2.29 \pm 0.03\text{b}$	$2.12 \pm 0.13\text{a}$	$2.16 \pm 0.11\text{a}$
AM-D	$0.55 \pm 0.01\text{c}$	$0.55 \pm 0.01\text{b}$		$1.98 \pm 0.02\text{c}$	$1.98 \pm 0.02\text{b}$	
CK-D	$0.16 \pm 0.03\text{c}$	$0.16 \pm 0.04\text{b}$		$1.69 \pm 0.04\text{d}$	$1.69 \pm 0.04\text{c}$	

AM-P, Proximal roots of *A. fruticosa* inoculated with AM fungus; CK-P, Proximal roots of *A. fruticosa* without AM fungi inoculation; AM-D, Distal roots of *A. fruticosa* inoculated with AM fungus; CK-D, Distal roots of *A. fruticosa* without AM fungi inoculation. Values with different letter are significantly different at $p < 0.05$.

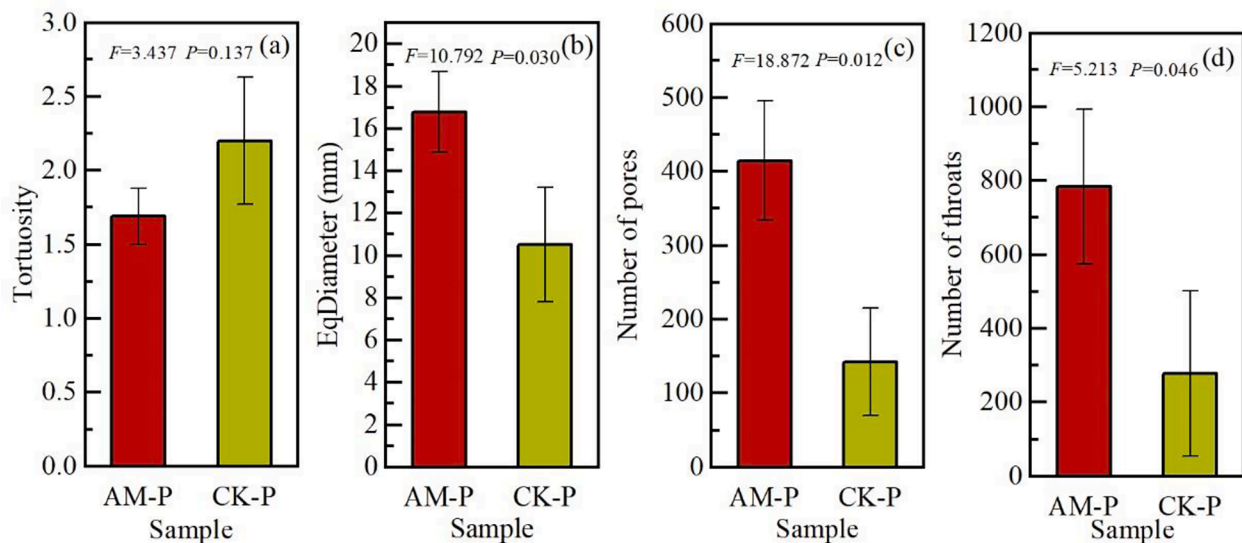


Fig. 4. AM-P, Proximal roots of *A. fruticosa* inoculated with AM fungi; CK-P, Proximal roots of *A. fruticosa* without AM fungi inoculation. a) Tortuosity of connected pores, b) Equivalent diameter of the connected pore, c) The number of pores in a connected pore skeleton, d) The number of throats in a connected pore skeleton. If $P < 0.05$, there is a significant difference between treatments.

4. Discussion

4.1. Effect of AM fungi inoculation on the root system

In this study, we explored the fascinating dynamics of AM fungi colonization, delving into its reliance on factors such as nutrient availability, host specificity, plant genotype, and the intriguing variations in nutrient uptake under different environmental conditions (Gosling et al., 2016; Yang et al., 2015a). Our findings revealed that inoculation with AM fungi had a significant impact, notably increasing both dry matter (DM) and HD at the proximal root of *A. fruticosa* (Fig. 6). These results underscored the establishment of a beneficial symbiotic relationship between AM fungi and the root system of *A. fruticosa*, particularly under the challenging conditions of compacted soil in the open-pit dump. Furthermore, artificial inoculation of AM fungi appeared to promote the growth and colonization of mycelia, shedding light on new avenues for enhancing plant growth and resilience in challenging environments.

To access more nutrients and water, roots need to expand their growth range (Hopmans and Bristow, 2002). However, in the compacted soil layer of the dump, roots must overcome the hard soil, which is a challenging process that limits plant growth (Kautz et al., 2013). Strock et al. (2022) study pointed out that root penetration of compacted soil requires higher carbon cost, so the root growth rate becomes slower. Xiong et al. (2020) also reached a similar conclusion in their study on the growth of maize in compacted soil: compacted soil can significantly reduce the root system of maize by 30 %. In this study, inoculation with AM fungi significantly increased the density characteristics of plant roots in the soil (RLD, RSD, RVD, and Tip number) (Fig. 5a ~ d), which was consistent with the findings of Berta et al. (2005) and Zheng et al. (2023). The tiny diameter of mycelium, typically only about 2 μm (Friese and Allen, 1985), enables it to penetrate soil pores that are unreachable by roots alone (Allen, 2007). This results in the formation of a vast mycelial network within the dump's soil, which facilitates plants in accessing a broader range of water and nutrients, thereby promoting plant growth and enhancing root system development (Chen et al., 2018). Concerning root morphological characteristics (Fig. 5E-h), inoculation with AM fungi significantly improved only root diameter (D) and RTD. This finding was consistent with the conclusions drawn by Cosme and Wurst (2013) and Hashem et al. (2019) in their studies. It suggests that under the compacted soil conditions of the open-pit dump, mycelium can, to some extent, replace the function of fine roots

(Pregitzer et al., 2002). Therefore, to secure a larger living space, roots must develop more coarse roots to penetrate the compacted soil resistance and expand (Whalley et al., 2008). Plants adapt to environmental changes by altering their own traits under stress (Orłowska et al., 2010). This study demonstrated that the artificial inoculation of AM fungi played a positive role in the adaptation of *A. fruticosa* to the arid and semi-arid conditions of the waste dump environment.

4.2. Effects of AM fungi inoculation on soil porosity

In this study, the CT scanning method was employed to describe soil pores, quantify the number of pores, pore geometry, and macroscopic pore parameters, and analyze the equivalent diameter, tortuosity, and pore throat number of connected pores. The results revealed that in the vicinity of *A. fruticosa*, the number of pores (Table 2) was significantly reduced ($P < 0.05$), which contradicted the research findings of Zhao et al. (2010). They suggest that as plant roots grow, the number of soil pores should continue to increase, eventually reaching an optimal number under forest soil conditions. However, in the compacted soil conditions of the dump, plant roots may preferentially seek areas with more pores to extend and grow. This preference leads to the fusion of multiple small pores in the soil into larger ones, consequently reducing the overall number of pores. To support this hypothesis, we utilized two sets of data.

Firstly, geometric parameters such as the average length, width, height, area, and volume of soil pores in the vicinity of *A. fruticosa* were significantly increased (Table 2) ($P < 0.05$). This suggested that plant roots increased the average size of soil pores, a conclusion consistent with the effect of plant roots on pores in natural soil (Liang et al., 2008; Wang KaiBo et al., 2007). Secondly, in the vicinity of *A. fruticosa*, soil macroporosity (Table 3) significantly increased ($P < 0.05$), a result similar to the research findings of Zhao et al. (2013). In the far end of the root system, there were no connected pores in the soil, while connected porosity accounted for 57 %~87 % of the total porosity in the vicinity of the root. In summary, plant roots played two crucial roles in the formation of the soil pore system in the dump: they enlarge pore size and connect individual pores.

Inoculation with AM fungi did not have a significant effect on the geometric size of soil pores (Table 2) ($P > 0.05$). Instead, it primarily affected the macro-porosity near the root (Table 3), particularly the connected porosity, which nearly doubled after inoculation with AM

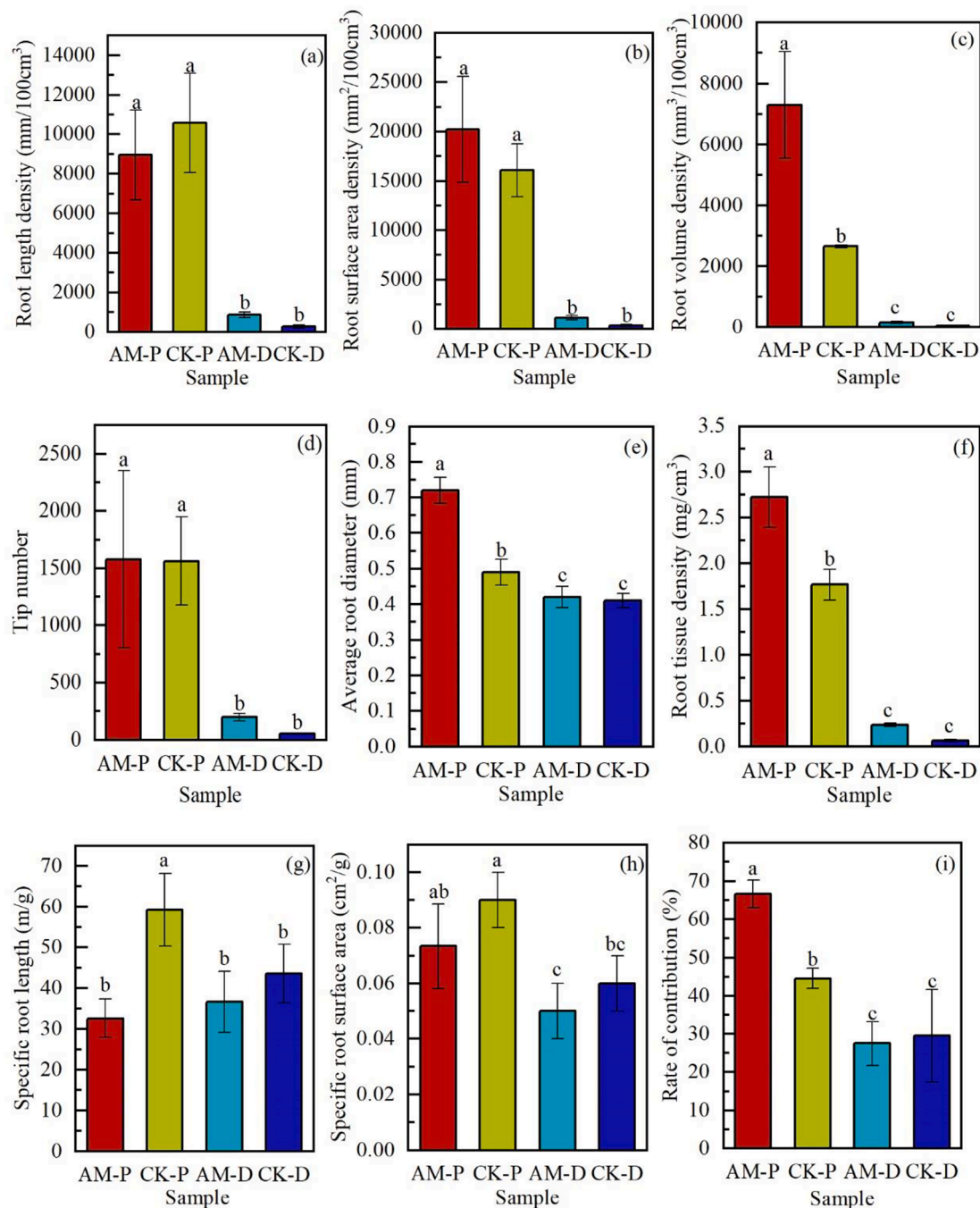


Fig. 5. AM-P, Proximal roots of *A. fruticosa* inoculated with AM fungus; CK-P, Proximal roots of *A. fruticosa* without AM fungi inoculation; AM-D, Distal roots of *A. fruticosa* inoculated with AM fungus; CK-D, Distal roots of *A. fruticosa* without AM fungi inoculation. Values with different letter are significantly different at $p < 0.05$.

fungi. The contribution rate of the AM-P root system to pores (Fig. 6i) was 66.68 %, significantly higher than that of other treatments ($P < 0.05$).

Partial least squares path analysis (Fig. 8) revealed that inoculated AM fungi indirectly affected soil macroporosity by influencing root morphological characteristics, a finding consistent with previous studies (Demenois et al., 2018; Morris et al., 2019; Veresoglou et al., 2012). The interaction between plant roots and mycorrhiza is a key driver of soil structure, with AM being particularly significant (Bardgett et al., 2014; Rillig and Mummey, 2006). Some studies have shown that the synergistic effect of leek root and AM fungi enhances soil porosity, which is several orders of magnitude larger than the pore volume generated by

leek roots alone (Milleret et al., 2009). In this study, inoculation with AM fungi significantly increased the equivalent diameter and throat number of connected soil pores. The increase in pore throats makes the soil pore system more complex, enhancing the mobility of soil water and gas (Koestel et al., 2018). This increased complexity attracts more roots and mycorrhiza to colonize the area, thus forming a positive cycle in soil ecology.

4.3. Effects of AM fungi inoculation on infiltration characteristics

In this study, Pearson correlation analysis (Table S1) revealed that HD and MC had an extremely significant positive correlation with soil

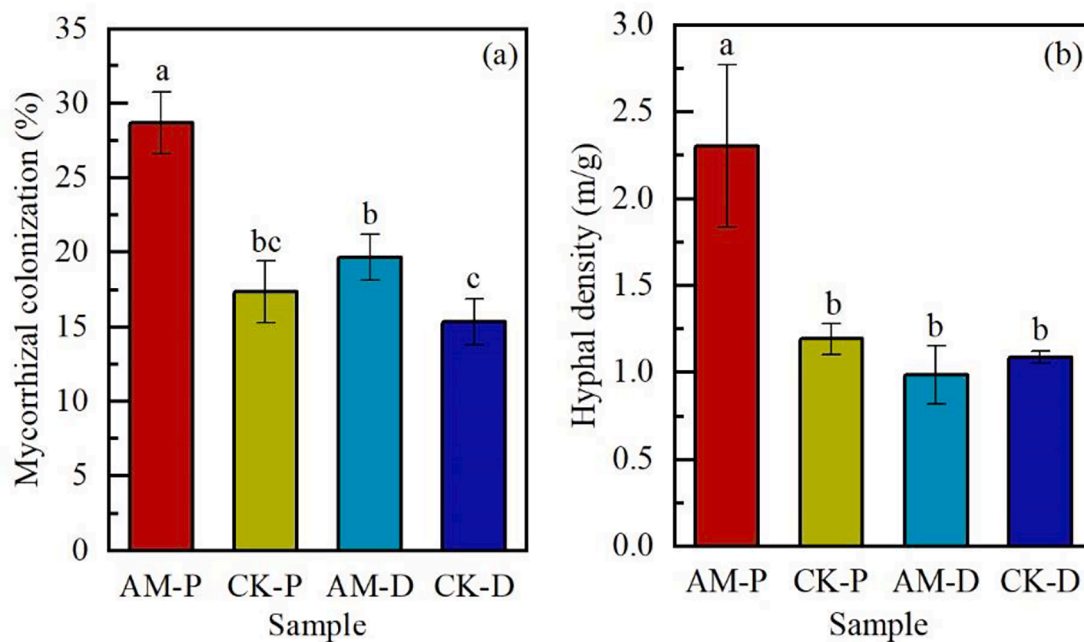


Fig. 6. AM-P, Proximal roots of *A. fruticosa* inoculated with AM fungus; CK-P, Proximal roots of *A. fruticosa* without AM fungi inoculation; AM-D, Distal roots of *A. fruticosa* inoculated with AM fungus; CK-D, Distal roots of *A. fruticosa* without AM fungi inoculation. Values with different letter are significantly different at $p < 0.05$.

infiltration characteristics (IIR, T15IR, SIR, and AIR) ($P < 0.01$). However, no studies have proved that AM fungi directly affect soil porosity and infiltration. The partial least squares path analysis in this study also indicated (Fig. 8) that AM fungi indirectly affect soil macroporosity and infiltration characteristics by influencing plant root morphology.

Some studies have suggested that AM fungi can secrete a spharomycin-associated protein (GRSP), which acts as a “binder” of soil”, effectively increasing the stability of soil aggregates (Bardgett et al., 2014). Additionally, the entanglement and interpenetration of mycelia are conducive to aggregate formation (Rillig and Mummey, 2006). Although these two effects of AM fungi can indirectly promote the formation of soil macropores by influencing soil aggregates, in this study, AM fungi did not have a significant direct effect on soil pores and soil infiltration rate (Fig. 8). This might be because the reclamation time of the waste dump in the study area was only 4 years, and mycelium and its secretions might not have a significant impact in such a short period.

Soil infiltration characteristics (IIR, T15IR, SIR, and AIR), root density characteristics (RLD, RSD, RVD, and Tips), root morphology characteristics (RTD and D), and soil macroscopic pore parameters (pore fraction, number of pores, number of throats, and EqDiameter) showed a highly significant positive correlation ($P < 0.01$), which was consistent with previous research findings (Hu et al., 2016; Kan et al., 2020; Saravanathiiban et al., 2014; Smet et al., 2018). In this study, inoculating AM fungi indirectly affected soil macroporosity (number of pores) and soil infiltration characteristics (IIR, T15IR, SIR, and AIR) by influencing root morphology (RTD and D). This differed from the research conclusion of Zheng et al. (2023), who suggests that in the early stages of vegetation restoration, soil structure is poor, macro porosity is low, and the pore network is relatively simple (Schlüter et al., 2020).

The squeezing effect of root growth on soil leads to a reduction in soil porosity (Cui Zeng et al., 2019). Therefore, root morphological characteristics have a negative impact on soil porosity. In the context of soil compaction in an open-pit dump, we believe that the action of the root system on pore formation can be divided into three steps. First, the root system needs to penetrate the hard soil layer and extend around it. Then, the roots search for small pores in the soil and connect them. Finally, the connected pores continue to expand, eventually forming large pores. MC and mycelium play crucial roles in this process. Mycelium can pass

through smaller pores to obtain nutrients and water for the root system over longer distances. Once the root system obtains sufficient nutrients, it can increase its diameter to have enough force to penetrate the soil resistance (Whalley et al., 2008). Additionally, AM fungi can induce changes in root chemistry, affecting the turnover of underground roots, thereby altering the decomposition rate and products of root decomposition, which ultimately form new and larger pores in the soil (Langley and Hungate, 2003).

In this study, under the influence of AM fungi, changes in root morphological characteristics not only indirectly affected soil infiltration characteristics through soil pores but also directly impacted soil infiltration characteristics (Fig. 8). This was because both the surface of living roots and the pores generated by dead roots are the main channels for soil water transport (Jačka et al., 2021; Li et al., 2021). Under the combined action of AM fungi and roots, the rate of development of the soil pore system in the dump was accelerated, and the increase in root diameter (D) and RTD became important factors affecting the soil infiltration process in the dump. Although many scholars have reached conclusions similar to ours, suggesting that an increase in root biomass and mean diameter will positively affect soil pore development and water infiltration (Archer et al., 2002; Leung et al., 2015; Zhang et al., 2016), we believe that different stages of root growth in different soil environments have varying effects on pores. The inoculation of *A. fruticosa* with AM fungi in the open-pit coal mine dump in arid and semi-arid areas undoubtedly promoted the rate of improvement of soil structure by roots, thus facilitating pore development and accelerating soil water infiltration. Inoculated AM fungi affect the agglomeration of plant rhizosphere soil, making it essential to further investigate the relationship between rhizosphere aggregates, soil pores and water infiltration. This study is crucial for understanding the mechanisms by which AM fungi enhance soil pore structure.

5. Conclusions

In arid and semi-arid regions, the humble *A. fruticosa* boasts a remarkable ability to rejuvenate even the most compacted soils found in open-pit dumps. Through a fascinating symbiotic relationship, inoculation with AM fungi sparks a burst of root growth, enhancing RLD,

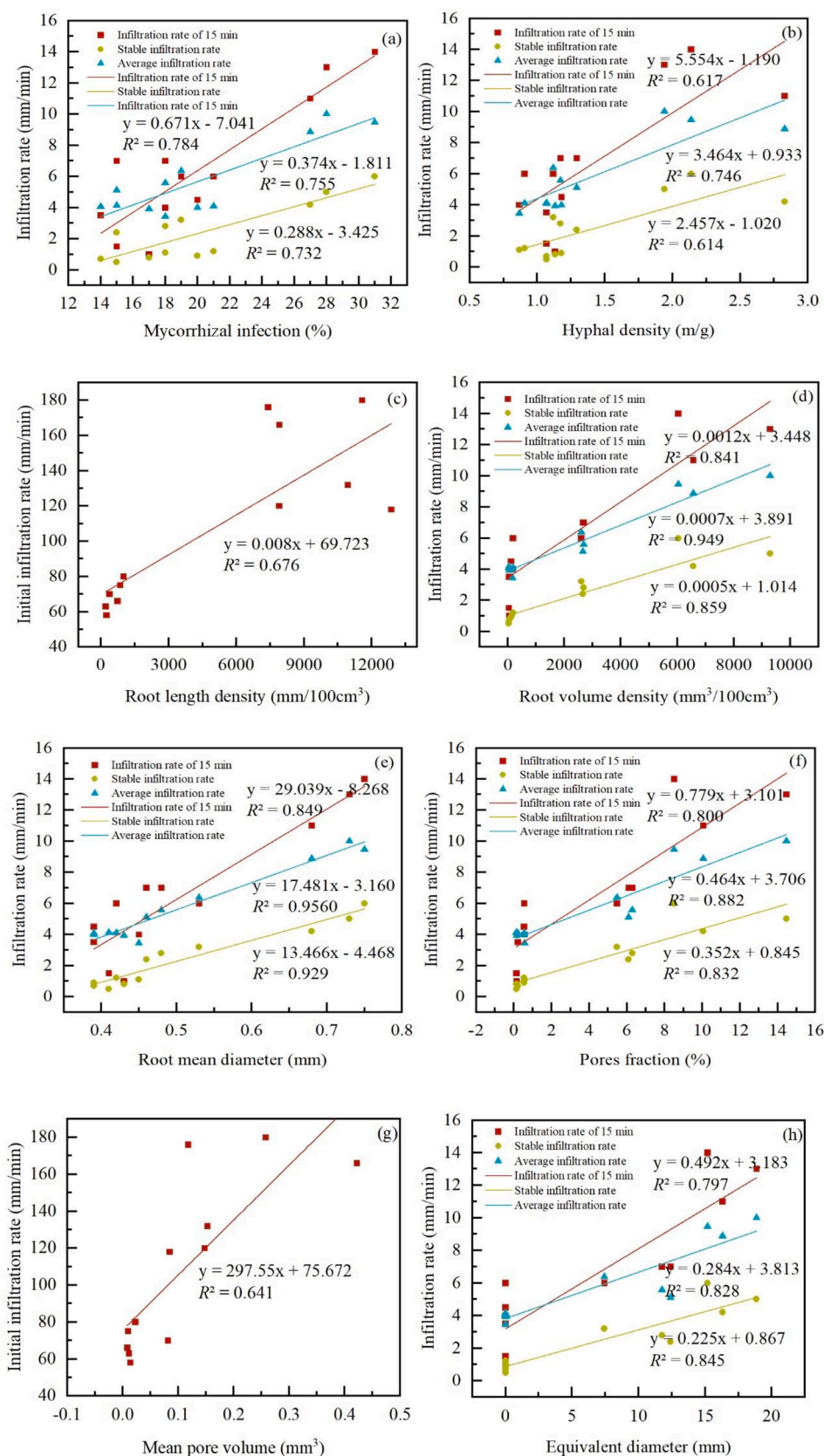


Fig. 7. The fitting relationship between infiltration characteristics and influencing factors.

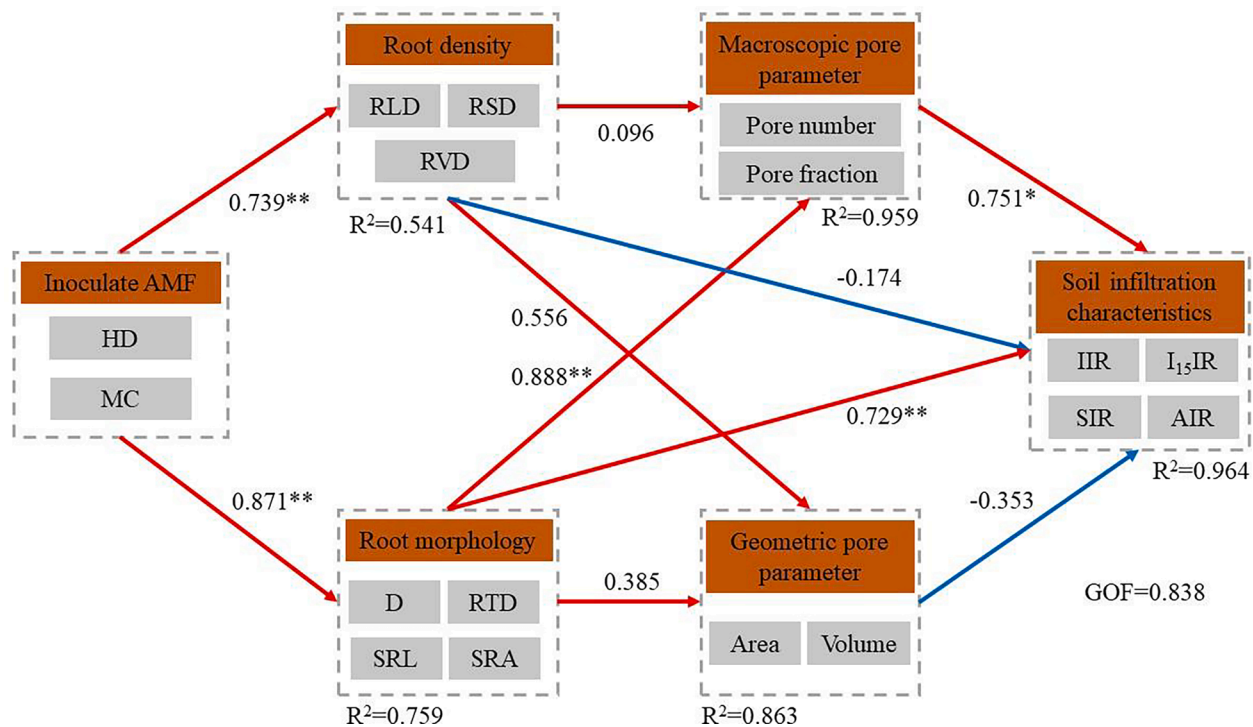


Fig. 8. Partial least squares path analysis of soil infiltration characteristics by AMF. Red lines indicate positive effects, whereas blue lines indicate negative effects. Numbers near arrows are standardized path coefficients. Significant values are indicated by * (P < 0.05), ** (P < 0.01), and numbers near the boxes indicate the variance explained by the model (R²). GOF means goodness of fit. (n = 3).

surface area, and volume. This not only alters the root's appearance, favoring robust coarse roots but also sets off a transformative chain reaction in the soil. As these roots delve deeper, they create an intricate web of interconnected pores within the compacted soil, enriching its structure. AM fungi play a pivotal role, accelerating this process significantly. By indirectly influencing macroscopic pore parameters, they further optimize soil infiltration capacity. These findings not only illuminate the intricate dance between AM fungi, root morphology, and soil structure but also hint at the vast potential of microbial technology in restoring ecological balance to open-pit waste sites across arid and semi-arid regions.

Author contributions

Yinli Bi and Xinpeng Du conceived and supervised the project. Du Xinpeng participated in the determination of infiltration performance and pore structure. Yinli Bi and Xinpeng Du wrote the manuscript. Lexuan Tian, Mingchao Li, Kejing Yin participated in the field experiment layout and index determination. All authors reviewed and approved the final manuscript.

CRediT authorship contribution statement

Xinpeng Du: Writing – original draft, Project administration. **Yinli Bi:** Writing – original draft, Supervision. **Lexuan Tian:** Methodology, Investigation. **Mingchao Li:** Investigation, Data curation. **Kejing Yin:** Methodology, Investigation.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A. Supplementary material

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.catena.2024.108515>.

Data availability

Data will be made available on request.

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